

# Lecture 3.3: The Inverse Transformation Method

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## Motivation

A generator, once verified as in Lectures 3.1–3.2, supplies a stream of  $\text{Uniform}(0, 1)$  values. Almost nothing in statistics actually needs a  $\text{Uniform}(0, 1)$  variable for its own sake — what is wanted is a variable from some *target* distribution: Exponential waiting times, Normal errors, Binomial counts. The question this lecture answers is how to convert one uniform stream into a stream from any prescribed distribution, using only the target’s cumulative distribution function.

## Intuition

Picture the graph of a cumulative distribution function  $F_X$ , rising from 0 to 1 as  $x$  increases. Choose a height  $u$  between 0 and 1 on the vertical axis, draw a horizontal line across to where it meets the curve, and then drop straight down to the horizontal axis: the point reached is a value  $x$  with  $F_X(x) = u$ . If the height  $u$  is itself chosen uniformly at random on  $(0, 1)$ , then the horizontal position where the line meets the curve is *not* uniform — it is distributed exactly as  $X$  itself. This is because  $F_X$  stretches out the regions where  $X$ ’s density is small and compresses the regions where it is large, so that a uniform sweep up the vertical axis reproduces  $X$ ’s own probability weighting on the horizontal axis. Generating  $X$  therefore reduces to two steps: draw a uniform height  $U$ , and read off the corresponding position on the horizontal axis — which is precisely  $F_X^{-1}(U)$ .

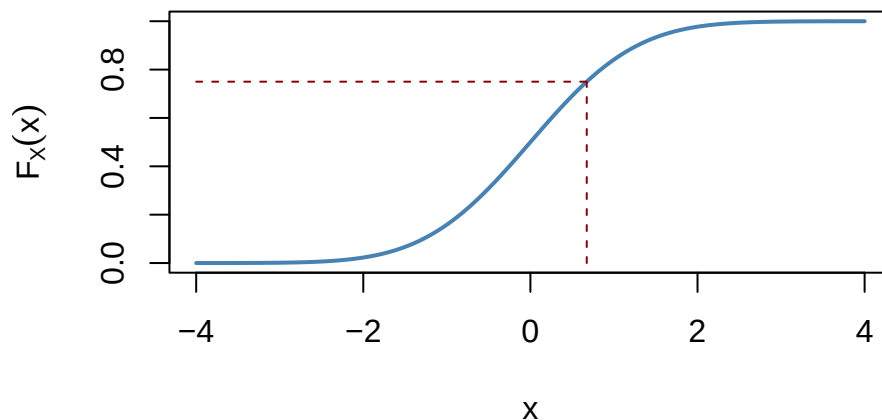


Figure 1: Reading the inverse cdf: a draw  $u = 0.75$  on the vertical axis maps, through  $F$ , to the simulated value  $x$ .

## The Quantile Function

The picture above is unambiguous whenever  $F_X$  is continuous and strictly increasing, since then it has an ordinary inverse function. The definition below extends this idea so that it makes sense even when  $F_X$  has jumps (as for a discrete random variable) or flat stretches, situations in which no ordinary inverse function exists.

**Definition 3.1.** For a cdf  $F_X$ , the *quantile function* (generalized inverse) is

$$F_X^{-1}(u) = \inf\{x \in \mathbb{R} : F_X(x) \geq u\}, \quad 0 < u < 1,$$

that is, the smallest  $x$  at which the cdf has climbed up to at least height  $u$ . It is non-decreasing (climbing  $F_X$  higher never requires a smaller  $x$ ) and left-continuous, and it satisfies the **compatibility relation**

$$F_X^{-1}(u) \leq x \iff u \leq F_X(x), \quad x \in \mathbb{R}, \quad u \in (0, 1),$$

which holds regardless of whether  $F_X$  is continuous, discrete, or a mixture of both, and is the single algebraic fact both theorems below are built from.

**Numerical illustration.** For  $X \sim \text{Bernoulli}(0.4)$ ,  $F_X(0) = 0.6$  and  $F_X(1) = 1$ , so  $F_X^{-1}(0.3) = 0$  and  $F_X^{-1}(0.8) = 1$  (the smallest  $x$  at which  $F_X$  reaches at least that height). Since  $0.3 < 0.8$  gives  $F_X^{-1}(0.3) = 0 \leq 1 = F_X^{-1}(0.8)$ , this concrete pair already confirms the non-decreasing property claimed above.

## Two Theorems

**Theorem 3.1 (Probability Integral Transformation).** If  $X$  is continuous with cdf  $F_X$ , then  $U = F_X(X) \sim \text{Uniform}(0, 1)$ .

*Proof.* Fix  $u \in (0, 1)$ . Using the compatibility relation with the roles of  $x$  and  $u$  read appropriately,

$$P(U \leq u) = P(F_X(X) \leq u) = P(X \leq F_X^{-1}(u)) = F_X(F_X^{-1}(u)) = u,$$

where the second equality is the compatibility relation itself, and the final equality holds because  $F_X$  is continuous, so applying  $F_X$  to its own generalized inverse returns the original argument exactly. Since  $P(U \leq u) = u$  for every  $u \in (0, 1)$ , this is precisely the  $\text{Uniform}(0, 1)$  cdf, so  $U \sim \text{Uniform}(0, 1)$ . ■

This first theorem explains *why* the method works, but it runs in the wrong direction for simulation — it starts from  $X$  and produces a uniform variable, whereas generation needs to start from a uniform variable and produce  $X$ . The next theorem is the converse, and is the one actually used to generate variates.

**Theorem 3.2 (Inverse Transformation).** If  $U \sim \text{Uniform}(0, 1)$  and  $F_X$  is any cdf, then  $X = F_X^{-1}(U)$  has cdf  $F_X$  — with no continuity assumption on  $F_X$  at all.

*Proof.* Fix  $x \in \mathbb{R}$ . By the compatibility relation,

$$P(X \leq x) = P(F_X^{-1}(U) \leq x) = P(U \leq F_X(x)) = F_X(x),$$

the last equality using  $U \sim \text{Uniform}(0, 1)$ , whose cdf is the identity function on  $(0, 1)$ , together with the fact that  $F_X(x)$  itself always lies in  $[0, 1]$ . Since  $P(X \leq x) = F_X(x)$  for *every*  $x$ , by definition  $X$  has cdf  $F_X$ . Notice that continuity of  $F_X$  was never used — the compatibility relation, and hence this proof, is valid whether  $F_X$  jumps, is flat, or is smooth. ■

## Applying the Theorem: Continuous and Discrete Cases

**Continuous case.** When  $F_X$  is continuous and strictly increasing,  $F_X^{-1}(u)$  is found by solving the single equation  $F_X(x) = u$  algebraically for  $x$ . For example, if  $X \sim \text{Exponential}(\lambda)$ , then  $F_X(x) = 1 - e^{-\lambda x}$  for  $x > 0$ ; setting this equal to  $u$  and solving gives

$$1 - e^{-\lambda x} = u \implies e^{-\lambda x} = 1 - u \implies F_X^{-1}(u) = -\frac{1}{\lambda} \log(1 - u).$$

**Worked example.** For the density  $f(x) = 3x^2$  on  $(0, 1)$ , integrating gives  $F_X(x) = x^3$ , so  $F_X^{-1}(u) = u^{1/3}$ . A uniform draw  $u = 0.729$  therefore produces  $x = 0.729^{1/3} = 0.9$ : this value of  $u$ , read as “72.9% of the probability mass lies at or below  $x$ ,” maps to the point  $x = 0.9$  at which exactly that much mass has accumulated, since  $F_X(0.9) = 0.9^3 = 0.729$  checks out exactly.

**Discrete case.** When  $X$  is discrete with support  $x_1 < x_2 < x_3 < \dots$ , there is generally no algebraic shortcut; the compatibility relation specializes to

$$F_X^{-1}(u) = x_i \quad \text{whenever} \quad F_X(x_{i-1}) < u \leq F_X(x_i),$$

so generating  $X$  means *searching* a table of cumulative probabilities for the first value at which the cumulative probability reaches or exceeds  $u$ , rather than inverting a formula.

**Worked example.** For  $X \sim \text{Bernoulli}(p = 0.4)$ ,  $F_X(0) = 1 - p = 0.6$  and  $F_X(1) = 1$ . A draw  $u = 0.55$  satisfies  $u \leq F_X(0) = 0.6$ , so  $X = 0$ ; a draw  $u = 0.70$  satisfies  $u > 0.6$ , so  $X = 1$ . In general, the sub-interval of  $(0, 1)$  mapped to each outcome has length exactly equal to that outcome’s probability — length 0.6 for the outcome 0 and length 0.4 for the outcome 1 — which is exactly why the method reproduces the correct distribution. Devroye (1986) catalogues quantile functions, in both closed and searchable form, for essentially every standard discrete and continuous distribution.

## References

Devroye, L. (1986). *Non-uniform random variate generation*. Springer-Verlag.